

Unit: Physical Quantities

Basanta Acharya, Physics Teacher



SUBSCRIBE



Basanta Acharya

Mastering the
alphabet & grammar
of physics! –
100% complete
notes.



Physical Quantities & Dimensions

Definitions

Physical Quantities: The quantities that can be measured directly or indirectly and in terms of which the laws of physics can be expressed are called physical quantities.

For examples: length, mass, density, area, etc.

Dimension: Dimension can be defined as the power or index to which the fundamental quantities are raised to express a physical quantity.

Rules:

- ★ Only fundamental quantities should be included while writing dimension.
- ★ Big brackets and capital initial letters should be used.
- ★ Numerical constants like 1, 2, 3, ..., π , e , etc. and angles like θ° , π^c , 180° , etc. have no dimension.

Dimension of Fundamental Quantities

Only seven fundamental quantities can be used to write dimension.

Mass,
Length,
and Time



The Big Three!
You'll use these
99% of the time
in mechanics.

S.N.	Fundamental Quantities	SI Unit	Dimension
1.	Mass	kg	[M]
2.	Length	m	[L]
3.	Time	s	[T]
4.	Luminous Intensity	cd	[Cd]
5.	Amount of Substance	mol	[Mol]
6.	Temperature	K	[K]
7.	Electric Current	A	[A]

Supplementary Quantities

Two quantities have been defined as supplementary quantities. These are neither fundamental nor derived. These are defined geometrically.

S.N.	Supplementary Quantities	Units
1.	Plane angle (θ)	Radian (rad)
2.	Solid angle (Ω)	Steradian (sterad) or (str)

Supplementary quantities are dimensionless quantities.

Dimension of Derived Quantities

a. Speed

$$[\text{Speed}] = \frac{[\text{distance}]}{[\text{Time}]} = [L T^{-1}]$$

b. Acceleration

$$\begin{aligned} [\text{Acceleration}] &= \frac{[\text{change in velocity}]}{[\text{Time}]} \\ &= \frac{[L T^{-1}]}{[T]} = [L T^{-2}] \end{aligned}$$

c. Force

$$\begin{aligned} [\text{Force}] &= [\text{Mass}] \times [\text{acceleration}] \\ &= [M] \times [L T^{-2}] \\ &= [M L T^{-2}] \end{aligned}$$

d. Pressure

$$\begin{aligned} [\text{Pressure}] &= \frac{[\text{Force}]}{[\text{Area}]} = \frac{[M L T^{-2}]}{[L^2]} \\ &= [M L^{-1} T^{-2}] \end{aligned}$$

Direct substitution!

e. Work Done

$$[\text{Work Done}] = [\text{Force}] \times [\text{displacement}]$$

$$= [M L T^{-2}] \times [L] = [M L^2 T^{-2}]$$

f. Density

$$[\text{Density}] = \frac{[\text{Mass}]}{[\text{Volume}]} = \frac{[M]}{[L^3]} = [M L^{-3} T^0]$$

g. Torque

$$[\text{Torque}] = [\text{radius}] \times [\text{Force}]$$

$$= [L] \times [M L T^{-2}]$$

$$= [M L^2 T^{-2}]$$

Look closely!
Different physical
concepts, but
EXACTLY the
same dimensions.

Principle of Homogeneity

The dimension of left hand side of any physical equation must be equal to the right hand side of that physical equation.

For example:

$$F = ma$$

$$\begin{aligned}[MLT^{-2}] &= [M] \times [LT^{-2}] \\ &= [MLT^{-2}]\end{aligned}$$

Time Period of Pendulum

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Dimension of LHS is: $[T]$

$$\begin{aligned}\text{Dimension of RHS is: } &\sqrt{\frac{[L]}{[LT^{-2}]}} \\ &= \sqrt{\frac{[LT^2]}{[L]}} \\ &= \sqrt{[T^2]} \\ &= [T]\end{aligned}$$

Applications of Dimensional Analysis

i) As a sanity check of Physical Equations

The two sides of any equation must be commensurable or have the same dimensions i.e. the equation must be dimensionally homogeneous.

The Setup

*To check the correctness of formula $v^2 = u^2 + 2as$, we proceed as:

The dimension of velocity $u = v = [M^0 L T^{-1}]$

The dimension of acceleration $(a) = [M^0 L T^{-2}]$

The dimension of displacement $(s) = [M^0 L T^0]$

The Proof

Now,

$$\text{LHS} = [M^0 L T^{-1}]^2 = [L^2 T^{-2}]$$

$$\begin{aligned}\text{RHS} &= [M^0 L T^{-1}]^2 + [L T^{-2}] \cdot [L] \\ &= [L^2 T^{-2}] + [L^2 T^{-2}] \\ &= [L^2 T^{-2}]\end{aligned}$$

In this case, $\text{LHS} = \text{RHS}$ [$\because$ constant 2 is dimensionless]

ii) To derive the relation between the physical quantities involved in a particular phenomenon.

Dimensional analysis can be used to derive a physical relationship between different physical quantities so that one can understand and characterize them.

For example, If the force $[F]$ depends on the mass of body $[m]$ moving in a particular path of radius (r) with velocity $[v]$, then the relation for centripetal force $(F) = \frac{m v^2}{r}$ can be derived by the method of dimension.

Let the centripetal force depend upon the mass of the body
i.e. $F \propto m^x$ --- (i)

Also, the centripetal force depends upon the velocity of the body
i.e. $F \propto v^y$ --- (ii)

And, the centripetal force depends upon the radius of the circle path
i.e. $F \propto r^z$ --- (iii)

Combining (i), (ii) and (iii), we get:

$$F \propto m^x v^y r^z$$

$$\text{or } F = k m^x v^y r^z \text{ --- (iv)}$$

Where k is a proportionality constant.



Deriving Centripetal Force (Continued)

From equation (iv):

$$\text{or } [M L T^{-2}] = k [M^x] [L T^{-1}]^y [L^z]$$

$$\text{or } [M L T^{-2}] = k [M^x] [L^{(y+z)}] [T^{-y}]$$

Equating powers of $[M]$, $[L]$ and $[T]$ on both sides:

$$x = 1$$

$$-y = -2 \quad \therefore y = 2$$

$$y + z = 1 \quad \therefore 2 + z = 1 \quad \therefore z = -1$$

Substituting the values of x , y and z on equation (iv),

$$F = \frac{k m v^2}{r}$$

Experimentally, value of k is found to be 1.

$$F = \frac{m v^2}{r}, \text{ This is the required expression.}$$

✓ Proved!

☆ Conversion of units of one system into another

This is based on the fact that the product of numerical value and the unit chosen of physical quantity always remain constant.

$$n_1 [U_1] = n_2 [U_2]$$

If M_1, L_1, T_1 be the fundamental units of mass, length and time respectively in first system of unit and M_2, L_2, T_2 be the fundamental units of them in second system of unit. Then physical quantity in first system of unit can be written as:

$$U_1 = n_1 [M_1^a L_1^b T_1^c]$$

Similarly, the physical quantity in second system of unit can be written as:

$$U_2 = n_2 [M_2^a L_2^b T_2^c]$$

The Master Formula

Since both measure the same physical quantity,

$$n_1 [U_1] = n_2 [U_2]$$

$$n_1 [M_1^a L_1^b T_1^c] = n_2 [M_2^a L_2^b T_2^c]$$

$$\therefore n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

To find dimension of constant

Dimensional analysis can be used to find the dimension of constants involved in different equations.

For example, From Newton's law of gravitation, the gravitational force acting in between two bodies of masses m_1 and m_2 separated by distance r can be written as:

$$F = \frac{Gm_1m_2}{r^2}, \text{ where } G \text{ is the gravitational constant.}$$

$$\therefore G = \frac{Fr^2}{m_1m_2}$$

Now, we have: $F = [M L T^{-2}]$, $m_1 = m_2 = [M^1 L^0 T^0]$, $r = [M^0 L^1 T^0]$

$$\begin{aligned}\therefore G &= \frac{[M L T^{-2}] \times [M^0 L^1 T^0]^2}{[M^1 L^0 T^0] \times [M^1 L^0 T^0]} \\ &= \frac{[M][L][T^{-2}] \times [L]^2}{[M]^2} \\ &= [M^{-1}][L^3][T^{-2}] \\ &= \underline{\underline{[M^{-1} L^3 T^{-2}]}}\end{aligned}$$

Limitations of Dimensional Analysis

- ✗ It gives no information about pure number, dimensionless constant.
- ✗ It gives no information about the dependence of a physical quantity in trigonometric, exponential and logarithmic functions.
- ✗ If a physical quantity of mechanics depends on more than three physical quantities then dimensional analysis can't be used to derive their relationship.
- ✗ The dimensional method is not applicable to derive formula having more than one term either in LHS or RHS.
- ✗ The dimensional method cannot differentiate different physical quantities having same dimension.
- ✗ The dimensional method cannot distinguish vector and scalar quantities.

Remember
Work and Torque
from earlier!



Least Count, Precision and Accuracy

Least Count

The smallest value which can be measured by a device.

Examples:

Normal scale = 1mm

Vernier caliper = 0.1mm

Micrometer screwgauge = 0.01mm

Precision

The degree of reproducibility of observed values is called precision of the measurement.

$$\text{Sensitivity of device} \propto \frac{1}{\text{Least count}}$$

$$\text{Sensitivity} \propto \frac{1}{\text{Range}}$$

Accuracy

The degree of closeness of observed values to the standard value is called accuracy of measurement.

Least count: Normal scale > vernier calliper > Screwgauge

Sensitivity: Normal scale < vernier calliper < screwgauge

Range: Normal scale > vernier calliper > screwgauge

Accuracy: Normal scale < vernier calliper < screwgauge

Significant Figures

The number of digits contained by a measured value which consists of the certain digit(s) along with an additional uncertain digit are called significant figures. It doesn't depend on system of units.

Significant Figures = certain digit(s) + uncertain digit

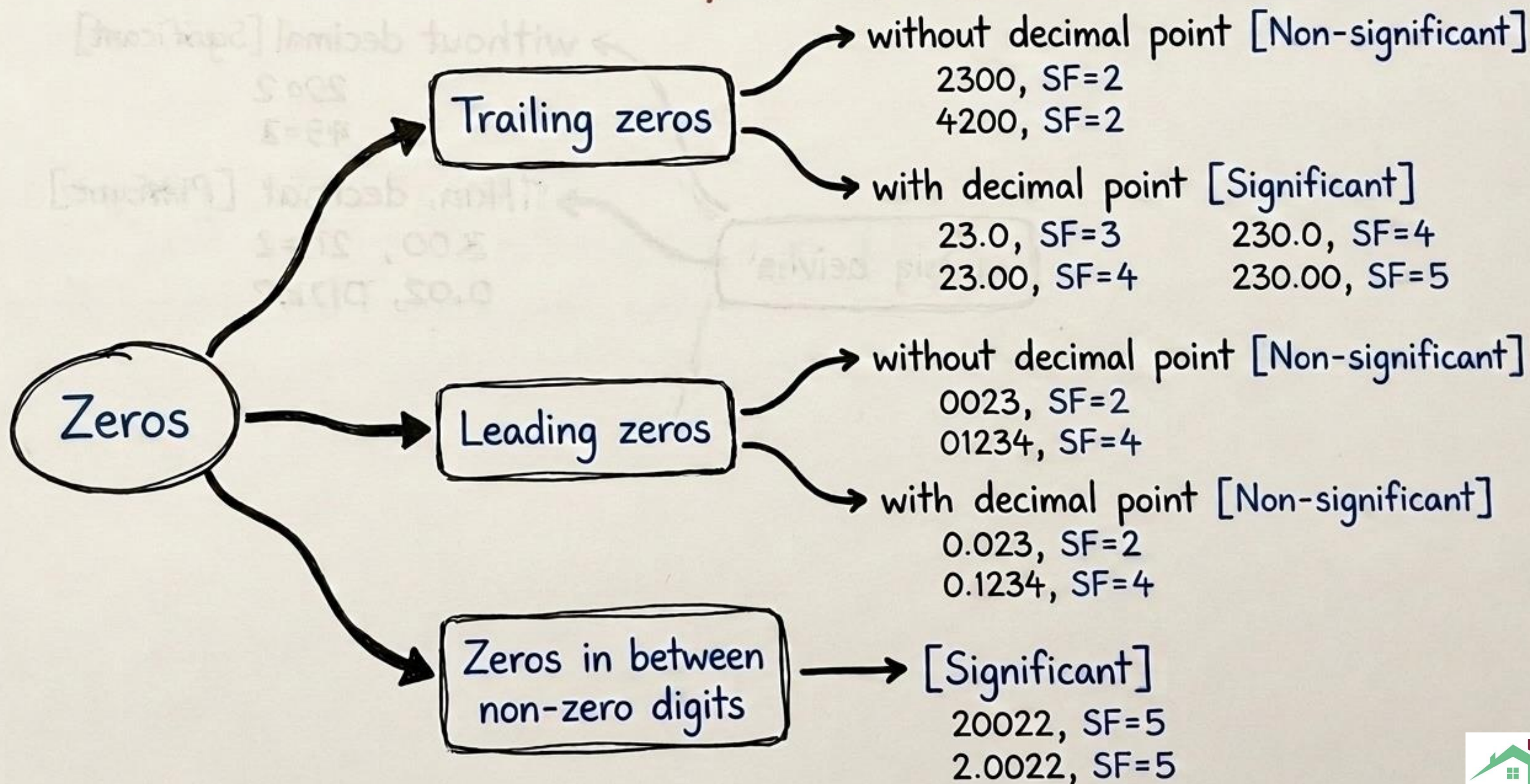
Rules for Significant Figures

- Non-zero digits [Significant]
a. 1.23, SF=3 | b. 3456, SF=4
- Power of 10 [Non-significant] 
 2.3×10^2 , SF=2 | 1.6×10^{-27} , SF=2
 9.1×10^{-31} , SF=2 | 1.6×10^{-27} , SF=2
 1.9×10^{30} , SF=2

Scientific notation makes identifying SFs easy—just look at the front number!



• Rule for Zeros (The Tricky Part!)



Rounding Off

The process of reshaping the value by dropping the digit(s) in such a way that the result obtained is least deviated from the original value is called rounding off.

1.27, SF=3
↪ 1.3, SF=2

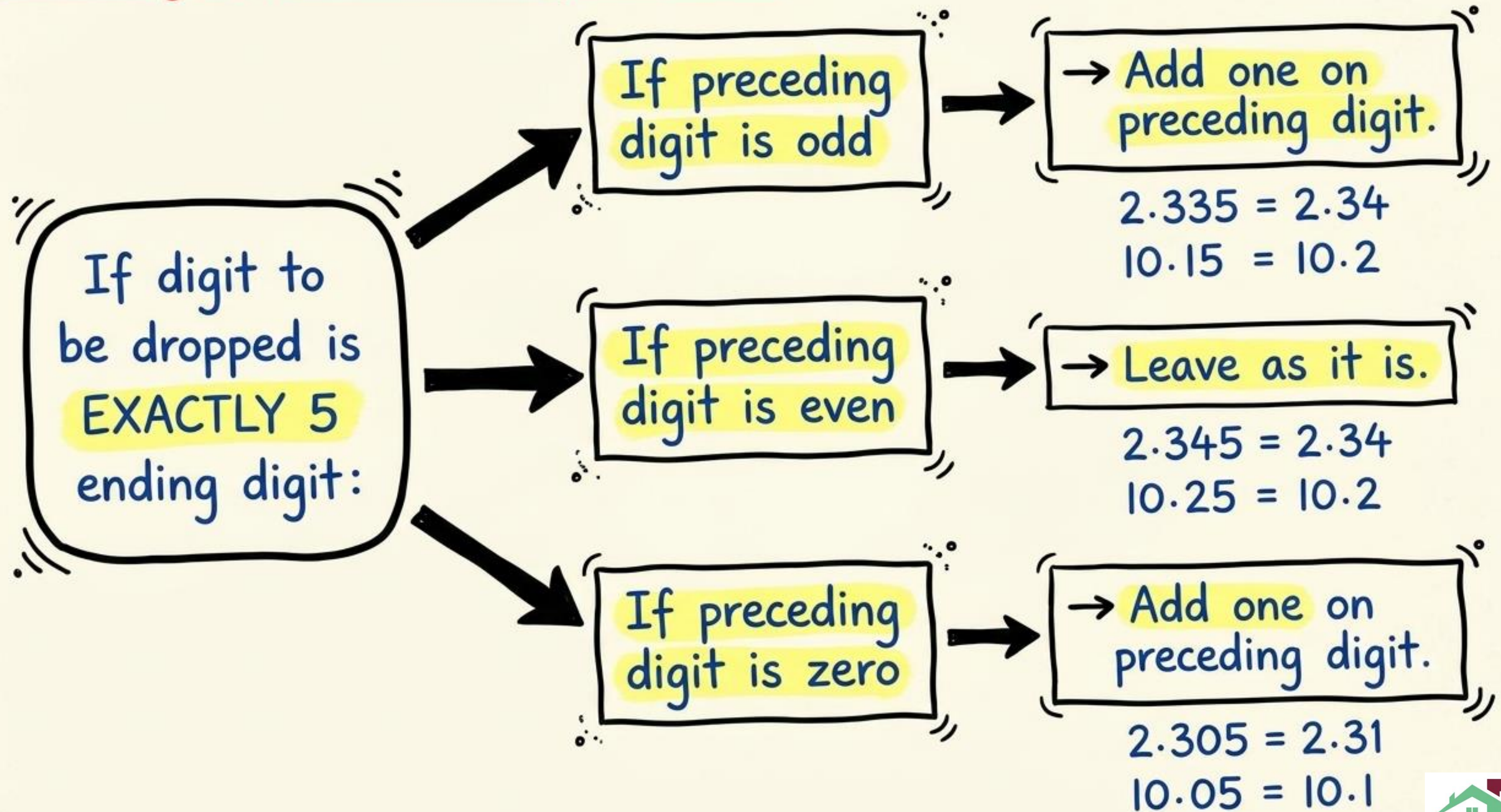
Rules if the digit to be dropped is:

→ Greater than 5
Add 1 in preceding digit.
 $2.347 = 2.35$

→ Lesser than 5
Leave preceding as it is.
 $2.342 = 2.34 \quad | \quad 0.271 = 0.27$

→ If 5 is followed by Non-zero digit
Add one on preceding digit.
 $10.052 = 10.1 \quad | \quad 10.151 = 10.2 \quad | \quad 10.253 = 10.3$

Rounding Off: The 'Exactly 5' Rules



Calculations with Significant Figures

For Addition and Subtraction	For Multiplication and Division
No. of digit(s) after decimal point in answer = No. of least digit(s) after decimal point of value given in question.	No. of significant figures in answer = No. of significant figures of value which is least accurate (in question).

Perform the indicated operations to correct significant figures.

a. $3.2 + 14.235$ $= 17.435$ $= 17.44$ $= \boxed{17.4}$	b. $27.482 - 27.4$ $= 0.082$ $= \boxed{0.1}$	c. 3.437×12 $= 41.244$ $= \boxed{41}$
d. $105 \div 5.5$ $= 19.09$ $= 19.1$ $= \boxed{19}$	e. $6.8 \times 10^4 + 3.1 \times 10^5$ $= 6.8 \times 10^4 + 31 \times 10^4$ $= 37.8 \times 10^4$ $= 3.78 \times 10^5 = \boxed{3.8 \times 10^5}$	

Applied Calculations (Word Problems)

Problem 1

What is the volume of box whose sides are 31.3cm, 28cm and 51.85cm?

Solution:

$$l = 31.3\text{cm}, b = 28\text{cm}, h = 51.85\text{cm}$$

Now,

$$\begin{aligned}\text{Volume (V)} &= l \times b \times h \\ &= 31.3 \times 28 \times 51.85 \\ &= 45441.34 \\ &= 45 \times 10^3 \text{ cm}^3\end{aligned}$$

Problem 2

If the length of cube is 7.203m. What is the TSA and Volume of cube to appropriate significant figures?

Solution:

$$l = 7.203\text{m}$$

$$\begin{aligned}\text{Now, TSA} &= 6l^2 \\ &= 6 \times (7.203)^2 \\ &= 311.299254 \text{ m}^2 \\ &= 311.3 \text{ m}^2\end{aligned}$$

$$\begin{aligned}\text{Now, Volume (V)} &= l^3 \\ &= (7.203)^3 \\ &= 373.7147544 \\ &= 373.7 \text{ m}^3\end{aligned}$$

The Cuboid Problem

The length, breadth and height of cuboid are 1.34m, 1.2m and 1m respectively. Find the TSA of cuboid to the correct significant figures.

Solution: $\ell = 1.34\text{m}$, $b = 1.2\text{m}$, $h = 1\text{m}$

Now,

$$\begin{aligned} A_1 &= \ell \times b = 1.34 \times 1.2 = 1.608 \text{ m}^2 = 1.6 \text{ m}^2 \\ A_2 &= b \times h = 1.2 \times 1 = 1.2 \text{ m}^2 = 1 \text{ m}^2 \\ A_3 &= \ell \times h = 1.34 \times 1 = 1.34 \text{ m}^2 = 1 \text{ m}^2 \end{aligned}$$

Now,

$$\begin{aligned} \text{TSA} &= 2(A_1 + A_2 + A_3) \\ &= 2(1.6 + 1 + 1) \\ &= 2 \times (3.6) \\ &= 2 \times 4 \\ &= \underline{\underline{8 \text{ m}^2}} \end{aligned}$$

Notice how strict SF rules in addition and multiplication drastically shape the final answer down to a single digit! Always round to least accurate inputs.